

# MUHAMMAD HAIDER ZAIDI

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🌐 [github.com/chefhaider](https://github.com/chefhaider) 🏠 Bavaria, Germany 🎂 21.06.2000

## Technical Skills

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**Programming:** Python, C++, SQL, Bash

**Tools/Frameworks:** TensorFlow, PyTorch, FastAPI, Docker, Kubernetes, AWS SageMaker, Flask, OpenAI API

**Competence:** Machine Learning, Statistical Modeling, Time-Series Forecasting, MLOps, Data Engineering, Data Visualization

## Experience

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### German National Metrology Institute (PTB)

March 2024 – Present

*Student Assistant Researcher*

*Berlin, DE*

- Designed and evaluated deep learning models for metrological applications, improving model performance and training efficiency.
- Developed physics-informed hybrid loss functions and novel architectures to enhance model accuracy and stability.
- Collaborated with PhD researchers to publish findings at the FLOMEKO conference.
- Built interactive dashboards for real-time data visualization and model performance monitoring.

### Folio3

March 2022 – Nov 2023

*Machine Learning Engineer*

*California, US (remote)*

- Achieved 94% accuracy and 5x speed improvement in vitiligo segmentation using advanced AI algorithms, deployed via Docker and FastAPI.
- Developed a real-time tennis foul detection system with TensorRT, reducing compute costs and processing time by 6x.
- Designed and managed MLOps pipelines on AWS SageMaker, addressing YOLO model data drift and maintaining Kubernetes infrastructure.
- Conducted educational sessions on machine learning and big data at universities, enhancing knowledge sharing and collaboration.
- Created POCs using YOLO, few-shot learning, and CNNs for client-specific use cases.

### Manda GmbH

March 2024 - June 2024

*Werkstudent Python Developer*

*Nürnberg, DE*

- Developed a full-stack web application for AI-based services using Flask, HTML, and CSS.
- Built an AI chatbot leveraging OpenAI API and large language model (LLM) frameworks for customer interaction.

### Freelance

July 2020 – March 2022

*Data Engineer*

*Remote*

- Developed web scrapers and ETL pipelines to process structured and unstructured data for analytics.
- Automated data ingestion and cleaning workflows, enabling efficient data processing for client projects.

## Projects

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### CAFA 6: Protein Function Prediction 2022

#### *Deep Learning Competition*

- Designed a weighted-probability ensemble integrating DPFunc and DeepGOPlus, leveraging AlphaFold for feature extraction.
- Enhanced model accuracy by 4% through advanced feature engineering and taxonomic bias mitigation.

### Razanlytics: Real-time Bitcoin Price Forecasting 2022

#### *Thesis Project*

- Developed a real-time Bitcoin price forecasting model using LSTMs, achieving a 92% prediction accuracy.

### Saaz: AI Self-Playing Chitrali Sitar 2023

#### *Signal Processing, Music Generation*

- Built the world's first AI-driven self-playing Chitrali Sitar, combining signal processing and music generation LLMs.
- Project featured on Coke Studio in a music video by Hassan Raheem, showcasing innovative AI-driven music technology.

### Hugging Face Open-Source

#### *Contributor*

- Co-authored Zero-shot Image Classification documentation and implementation page on the Hugging Face Hub.

## Education

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### Friedrich Alexander Universität Present

#### *Master of Science in Artificial Intelligence* Erlangen, DE

- Focused on machine learning, statistical modeling, and time-series forecasting for real-world applications.

### National University of Computer and Emerging Sciences Aug 2018 – June 2022

#### *Bachelor of Science in Computer Science* Karachi, PK

- Graduated with honors and recognized on the Dean's List of Honor.
- Specialized in data engineering, machine learning, and artificial intelligence.

## Certifications

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**PyTorch:** Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate  
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python